

Week 3 Introduction

Mathematics for Multidimensional Data

Working with Multidimensional Data in Python

-  **Video:** 3.7 Basic Plotting
2 min
-  **Reading:** Note on Matplotlib
10 min
-  **Practice Quiz:** Basic Plotting – Review Information
3 questions
-  **Reading:** Matplotlib Scatter Plot Documentation
20 min
-  **Video:** 3.7a Storing 2D Coordinates in a Single Data Structure
6 min
-  **Practice Quiz:** Storing 2D Coordinates – Review Information
2 questions
-  **Video:** 3.8 Multidimensional Mean
4 min
-  **Practice Quiz:** Multidimensional Mean – Review Information
2 questions
-  **Video:** 3.9 Adding Graphical Overlays
5 min
-  **Practice Quiz:** Adding Graphical Overlays – Review Information
3 questions
-  **Reading:** Matplotlib Patches Documentation
10 min



Note on Matplotlib

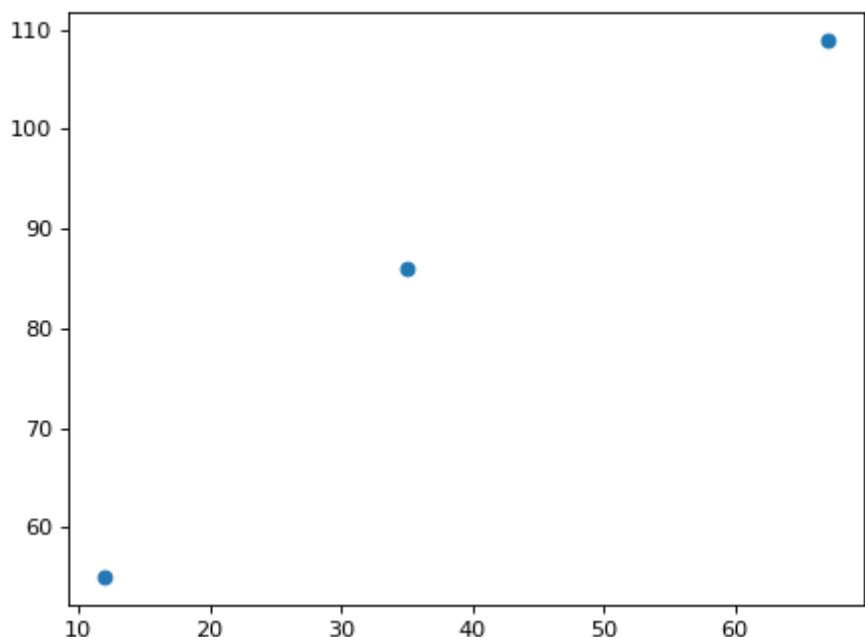
At around 1'20 in the previous video, you can see Matthew being puzzled when he runs his matplotlib code. This has been worrying some of you, and that this behaviour can be avoided entirely by invoking `plt.show()` in the code that creates the plot object and returns it, but it is the `plt.show()` function that does this. When presented with this object, the Jupyter notebook invokes an automatic display, sometimes not. It is recommended to use `plt.show()` for safety and clarity.

Here a code example you can try for yourself. The first one will show the automatic mechanism of the notebooks is not available), whereas the second one will show the automatic mechanism of the notebooks is not available).

```
1 xs = [12, 35, 67]
2 ys = [55, 86, 109]
3 plt.scatter(xs,ys)
```

```
<matplotlib.collections.PathCollection object at 0x7f41e27ef8d0>
```

```
1 xs = [12, 35, 67]
2 ys = [55, 86, 109]
3 plt.scatter(xs,ys)
4 plt.show()
```



-  **Video:** 3.10 Calculating the Distance to the Mean